Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/06/2009
PI Name:	Bingham, Gail E Ph.D.		
Project Title:	Optimization of Root Zone Substrates for Reduced Gravity Experiments		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Advanced human support technologies		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	UNIVERSITY	**Phone:**	435-797-4320
Organization Name:	Utah State University		
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PI Web Page:			
City:	North Logan	**State:**	UT
Zip Code:	84341	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2000 Space Biology 00-HEDS-01
Start Date:	12/01/2001	**End Date:**	02/28/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 2/28/2009 per annual report info (5/2008) NOTE: End date corrected to 02/21/2008 per PI (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jones, Scott (Utah State University)		
Grant/Contract No.:	NAG 9-1284		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Gas diffusion in all root zones is limited by the air-filled pore space within the plant growth substrate. Since root mass is an actively respiring tissue, most plants require sufficient oxygen diffusion in their root environment to support their root respiration rate. When root zone oxygen becomes limiting, plant growth suffers. At 1g (on earth) gravity pulls the water through the root zone draining the macro pore spaces, allowing oxygen to diffuse to the roots. In mg, water is held in the macro pore spaces and gas diffusion into and out of the root zone can become a limiting factor for plant growth at lower volumetric water contents. The objective of this research is to determine the parameters required to optimize the plant growth root environment for reduced gravity environments so that future researchers can confidently tailor their root zone environment to satisfy their experimental needs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Removing gravitational effects from the measurements of oxygen and carbon dioxide diffusion in plant substrates has provided us with significant new insight to how these processes function on earth. We have been able to improve the modeling of water-soil interactions with the work to date. At the completion of this effort, we will be able to provide greenhouse and other earthbound high intensity plant growers with the tools to develop improved substrates and water management practices. The Optimization of Root Zone Substrates (ORZS) experiment represents a highly successful interdisciplinary collaboration between Space Dynamics Laboratory (Logan, UT), The Institute for Medical and Biomedical Problems (Moscow, Russia), Utah State University (Logan, UT) and NASA. With the return of the hardware from the second flight to the International Space Station in April 2009, the ORZS program has successfully established the first detailed study of porous plant growth media in a long-term microgravity environment. Preliminary parabolic flight experiments were an important step in developing, testing, and understanding the challenges of performing fluid experiments in microgravity. Two critical properties of porous media were determined, the first being wetting and draining water-retention characteristics and the second, the corresponding oxygen diffusion transport coefficients. These data are critical for understanding water behavior in porous plant growth media and the associated transport of nutrients and oxygen to plant roots in the absence of gravity. Data from ground studies, parabolic flight experiments, and both ISS flight experiments have been analyzed and models describing these properties have been parameterized. While some lingering questions remain, the ORZS experiment provides solid evidence that water-retention characteristics are consistent between 1 g and microgravity. On the other hand, oxygen transport coefficients were significantly different in microgravity for the finer particle sized media tested. Further work is likely warranted to better understand a potential shift in the gas percolation threshold. This report summarizes the knowledge gained by Space Dynamics Laboratory (SDL) and Utah State University (USU) team members during planning, design, and construction of the ORZS hardware and development of the scientific experiments. This information has been or will be publicized in 23 presentations and in 10 peer-reviewed publications.
Task Progress:	The Optimization of Root Zone Substrates (ORZS) experiment represents a highly successful interdisciplinary collaboration between Space Dynamics Laboratory (Logan, UT), The Institute for Medical and Biomedical Problems (Moscow, Russia), Utah State University (Logan, UT) and NASA. With the return of the hardware from the second flight to the International Space Station in April 2009, the ORZS program has successfully established the first detailed study of porous plant growth media in a long-term microgravity environment. Preliminary parabolic flight experiments were an important step in developing, testing, and understanding the challenges of performing fluid experiments in microgravity. Two critical properties of porous media were determined, the first being wetting and draining water-retention characteristics and the second, the corresponding oxygen diffusion transport coefficients. These data are critical for understanding water behavior in porous plant growth media and the associated transport of nutrients and oxygen to plant roots in the absence of gravity. Data from ground studies, parabolic flight experiments, and both ISS flight experiments have been analyzed and models describing these properties have been parameterized. While some lingering questions remain, the ORZS experiment provides solid evidence that water-retention characteristics are consistent between 1 g and microgravity. On the other hand, oxygen transport coefficients were significantly different in microgravity for the finer particle sized media tested. Further work is likely warranted to better understand a potential shift in the gas percolation threshold. This report summarizes the knowledge gained by Space Dynamics Laboratory (SDL) and Utah State University (USU) team members during planning, design, and construction of the ORZS hardware and development of the scientific experiments. This information has been or will be publicized in 23 presentations and in 10 peer-reviewed publications.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Heinse R, Jones SB, Or D, Tuller M, Topham TS, Podolskiy IG, Bingham GE. "Challenges of Watering Plants in Space: Water Retention and Distribution---What Have We Learned?" 37th COSPAR Scientific Assembly, Montreal, Canada, 13-20 July 2008. 37th COSPAR Scientific Assembly, Montreal, Canada, 13-20 July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Heinse R, Jones SB, Or D, Topham TS, Podolskiy IG, Bingham GE. "Microgravity Implications of Water Distribution on Oxygen Diffusion Pathways in Unsaturated Porous Media." GSA-SSSA-ASA-CSSA-GCAGS-HGS 2008 Joint Annual Meeting, George R. Brown Convention Center, Houston, TX, Oct. 5-9, 2008. Agronomy Abstracts, ASA, Madison, WI, 2008. , Oct-2008
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